KNEE CONTROLLER

VARIANT: PRELIMINARY

2026-10-16

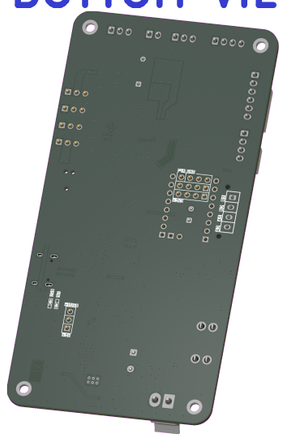
REV A

Page	Index	Page	Index	Page	Index	Page	Index
1	COVER PAGE	11	L1–L2 DRILL MAP	21	.....	31	.....
2	SYSTEM OVERVIEW	12	TOP TESTPOINTS	22	.....	32	.....
3	POWER MANAGEMENT	13	L1 COPPER	23	.....	33	.....
4	MICROCONTROLLER	14	L2 COPPER	24	.....	34	.....
5	MOTOR CONTROL AND IO	15	.....	25	.....	35	.....
6	ASSEMBLY OVERVIEW	16	.....	26	.....	36	.....
7	TOP ASSEMBLY	17	.....	27	.....	37	.....
8	BOTTOM ASSEMBLY	18	.....	28	.....	38	.....
9	TOP FABRICATION	19	.....	29	.....	39	.....
10	BOTTOM FABRICATION	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

MADE WITH LOVE

NOT FITTED COMPONENTS MARKED AS X

DRAFT – EARLY STAGE SCHEMATIC CAPTURE
PRELIMINARY – LATE STAGE SCHEMATIC CAPTURE
CHECKED – PCB LAYOUT CHECKED, CONTACT ENGINEER IF MISTAKE FOUND
RELEASED – READY FOR PRODUCTION

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

REV	DATE	DESCRIPTION	DRAWN	CHECK

SCALE NTS

SCHEMATIC

BOARDWRIGHT

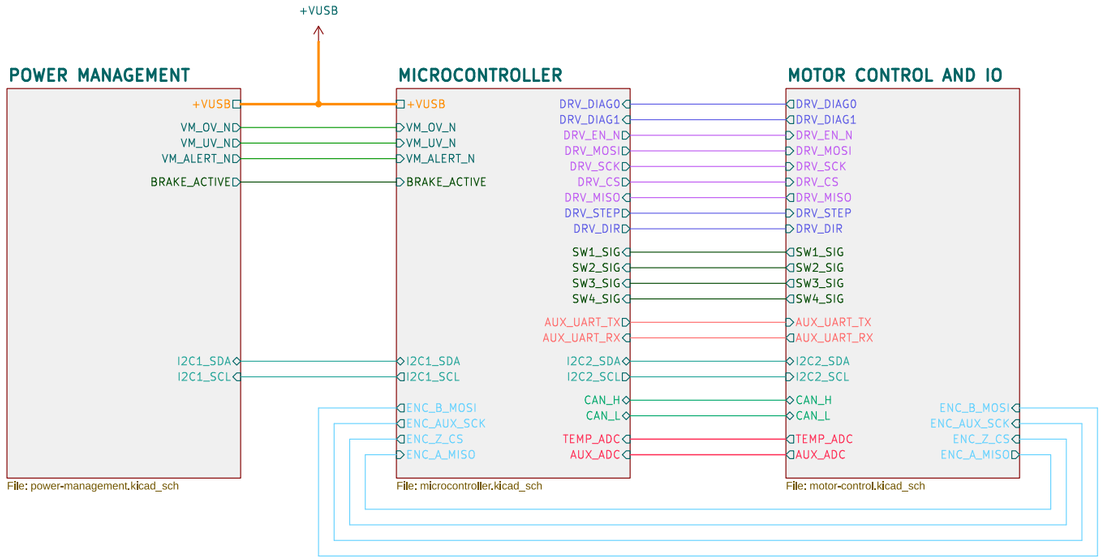
KNEE CONTROLLER

DRG. NO. 4840.703

REV A

A3

SHEET 1 OF 14



REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS

SCHEMATIC

A3

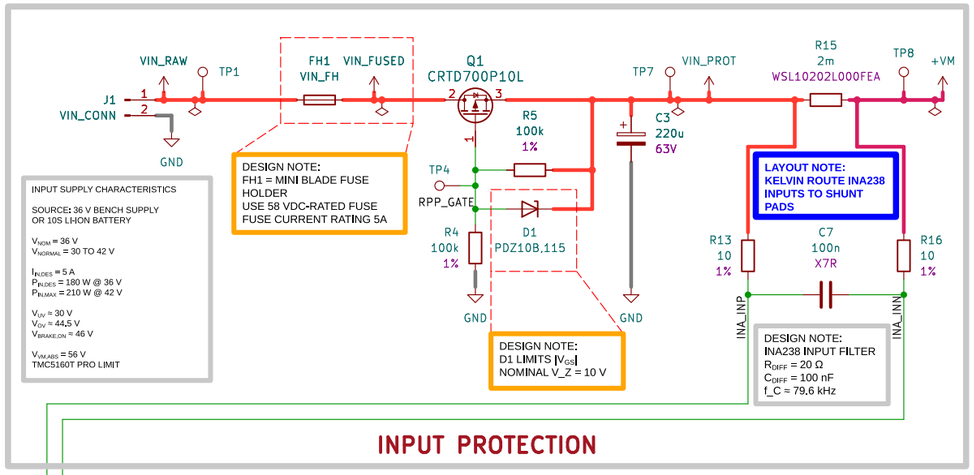
SHEET 2 OF 14

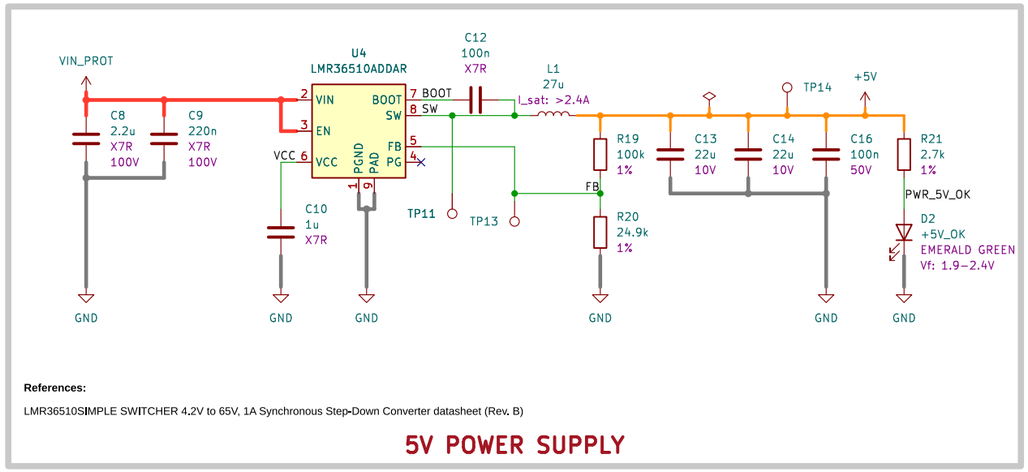
BOARDWRIGHT

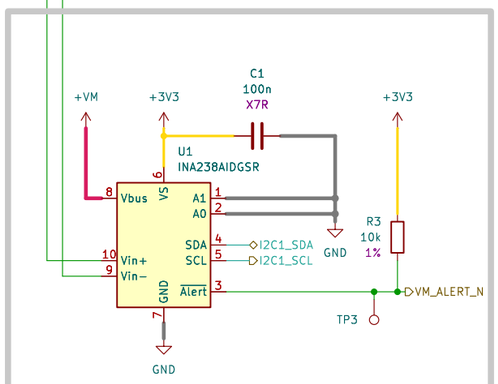
KNEE CONTROLLER

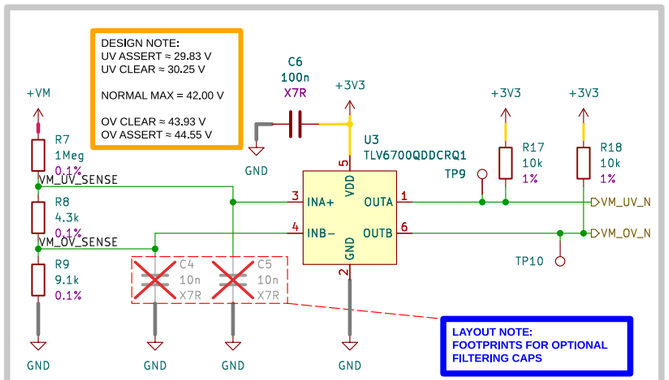
DRG. NO. 4840.703

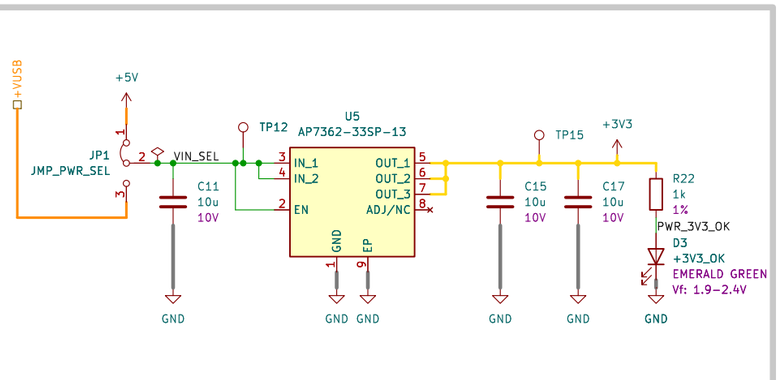
REV A

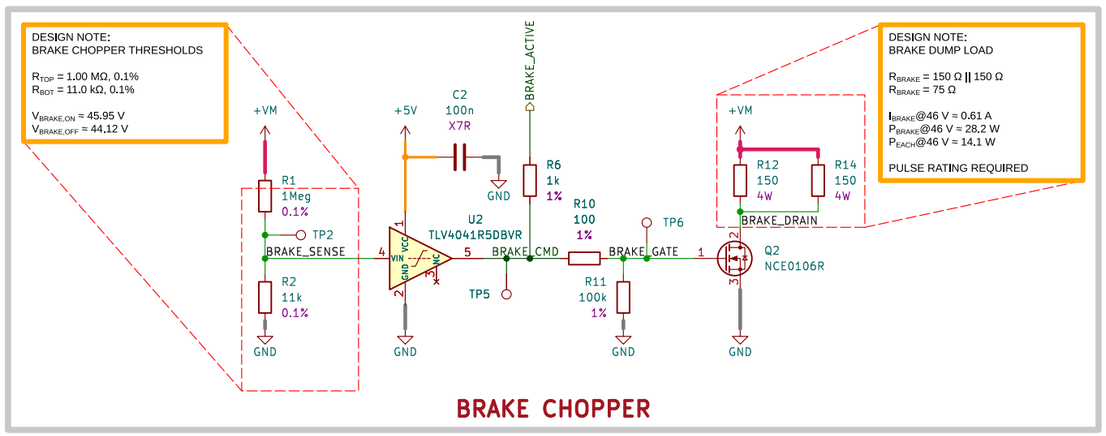












REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS

SCHEMATIC

BOARDWRIGHT

KNEE CONTROLLER

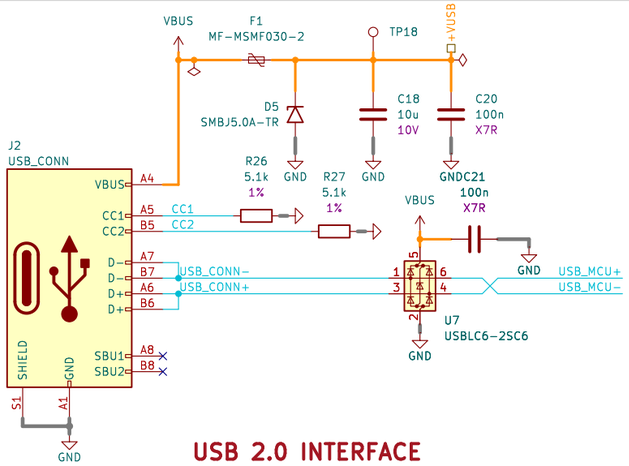
A3

SHEET 3 OF 14

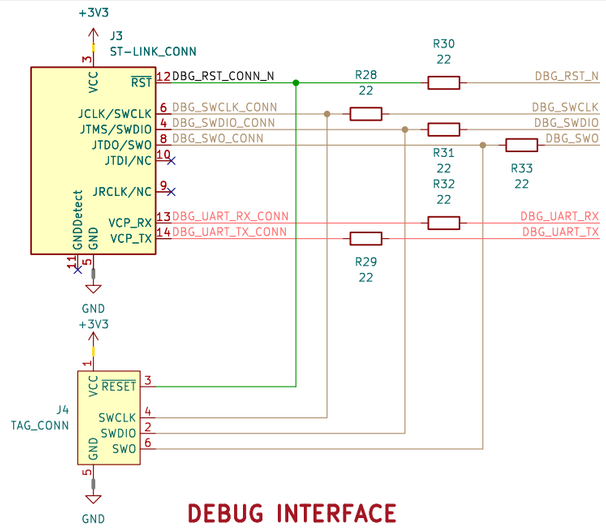
DRG. NO. 4840.703

REV A

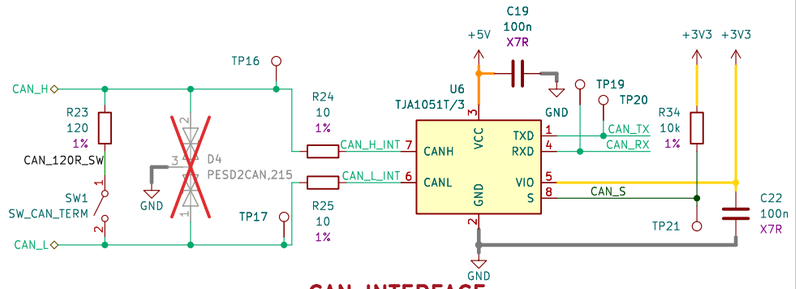
USB 2.0 INTERFACE



DEBUG INTERFACE



CAN INTERFACE



CLOCK SOURCES

DESIGN NOTE:
CRYSTAL LOAD CAPACITORS

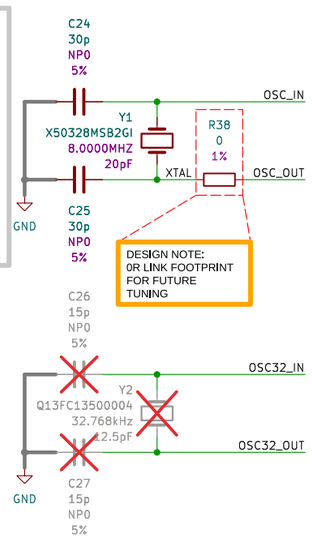
CL = (C1*C2)/(C1+C2) + CS

FOR C1 = C2 = C:
C = 2*(CL - CS)

HSE (Y1):
CL = 20 pF
CS = 10 pF (MCU + PCB EST.)
C24 = C27 = 20 pF
USE C0G/NP0

LSE (Y2):
CL = 12,5 pF
CS = 5 pF (INITIAL EST.)
C33 = C34 = 15 pF
USE C0G/NP0

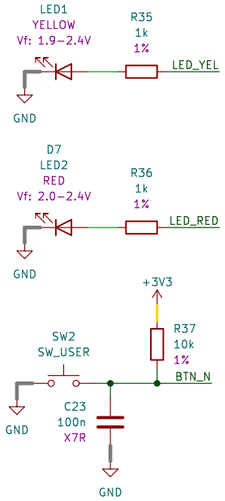
VALUES ARE INITIAL DESIGN VALUES.
VERIFY/TUNE ON FINAL PCB IF
FREQUENCY ACCURACY REQUIRES.



References:

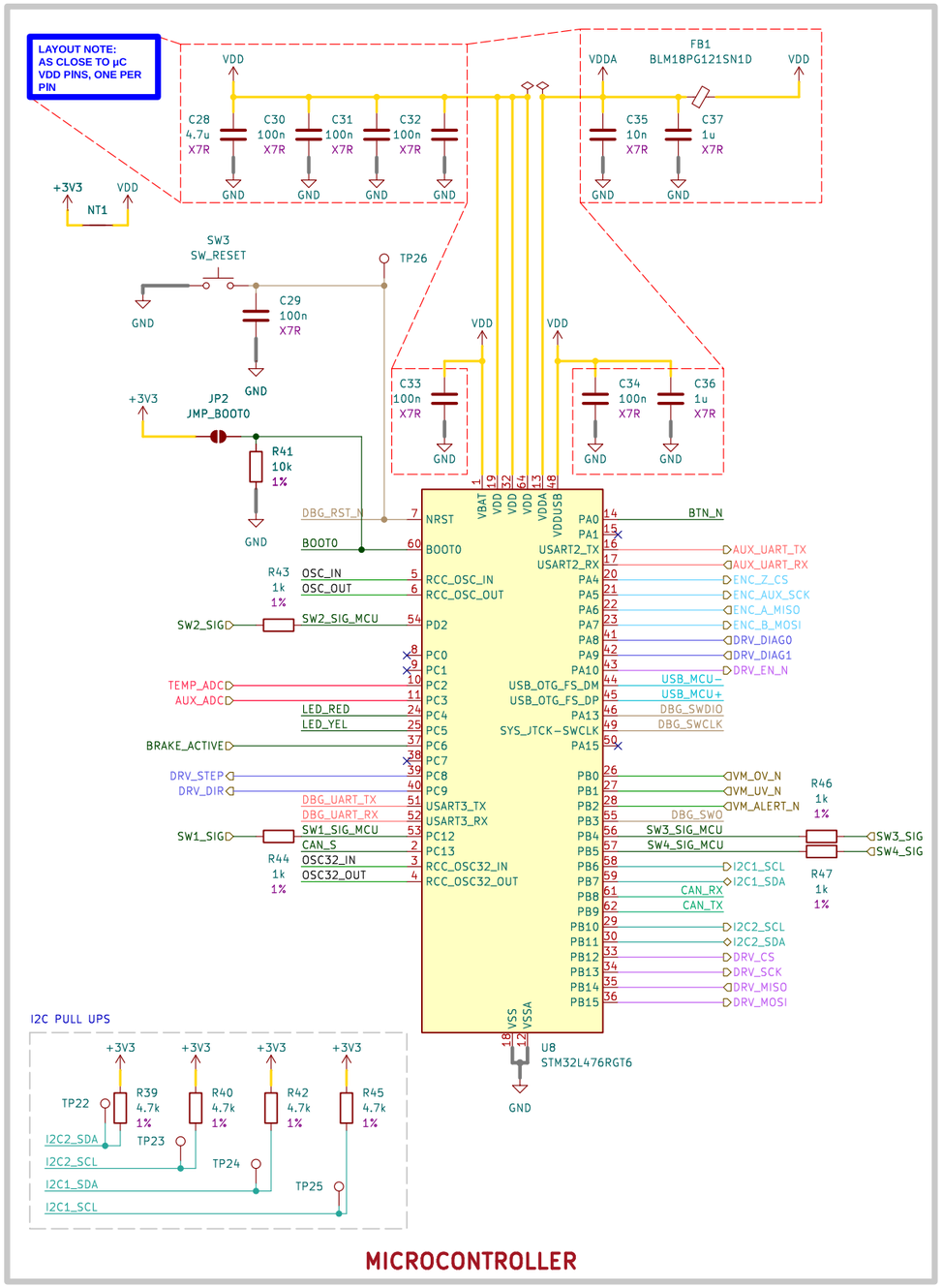
AN2867 - Guidelines for oscillator design on STM8AF/AL/S and STM32 MCUs/MPUs

USER INTERFACE



MICROCONTROLLER

LAYOUT NOTE:
AS CLOSE TO µC
VDD PINS, ONE PER
PIN



I2C PULL UPS

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

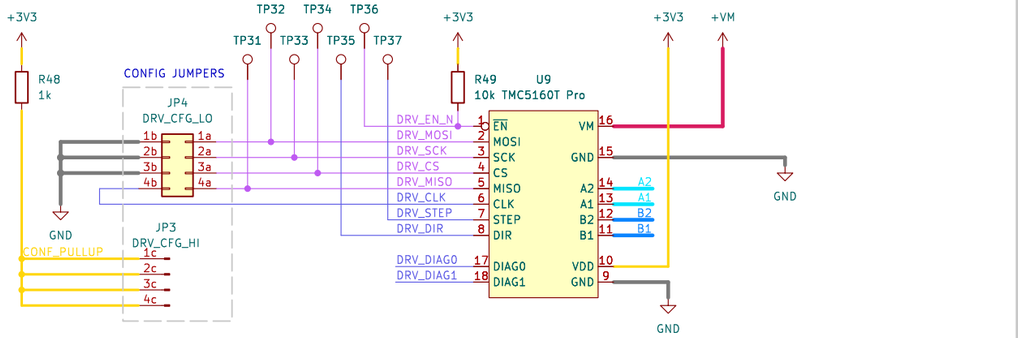
BOARDWRIGHT

SCHEMATIC

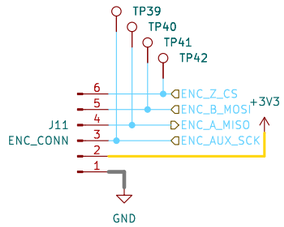
KNEE CONTROLLER

SCALE	DRG. NO.	REV
NTS	4840.703	A
A3	SHEET 4 OF 14	

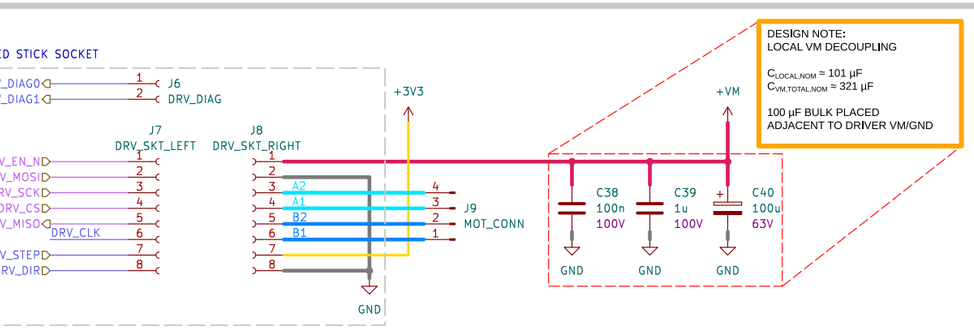
REV	DATE	DESCRIPTION	DRAWN	CHECK



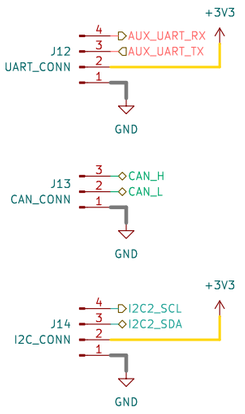
MOTOR DRIVER



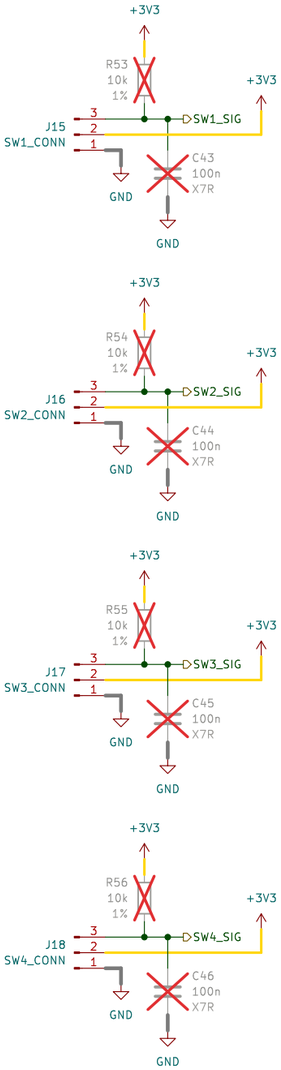
ENCODER INTERFACE



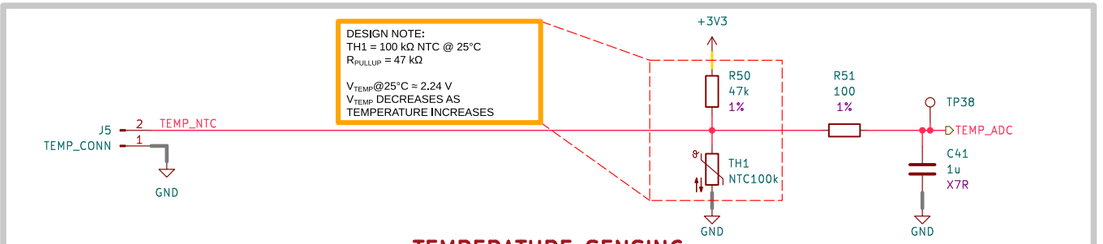
MOTOR INTERFACE



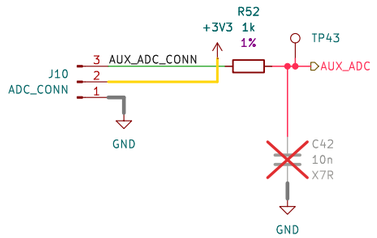
COMMUNICATION INTERFACES



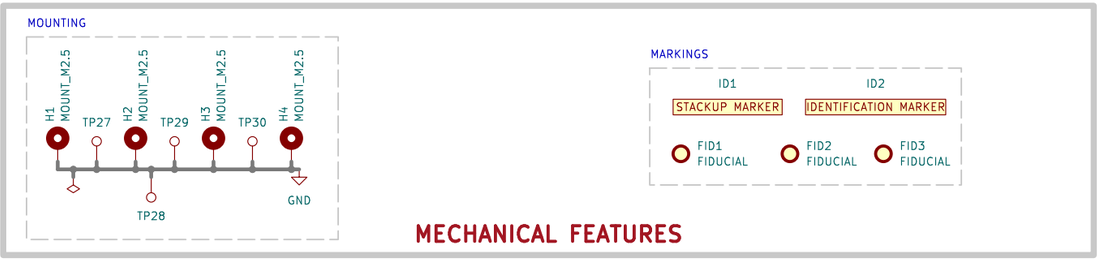
DIGITAL INPUTS



TEMPERATURE SENSING



ANALOG INPUT



MECHANICAL FEATURES

REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS

SCHEMATIC

BOARDWRIGHT

KNEE CONTROLLER

DRG. NO. 4840.703

REV A

A3

SHEET 5 OF 14